DO YOU KNOW?

Thiruvakkarai fossil wood park (Villupuram district, Tamil Nadu): 2 million years ago tree trunks that got buried along the river, incourse of time the organic matter was replaced by silica and was fossilized. They retained their color, shape and texture and was converted into solid rocks. The annular rings, the texture, colors of the layers, nodes and every properties of plants are still visible.



19.6 Ethnobotany

Ethnobotany is the **study of a region's plants** and their **practical uses** through the **traditional knowledge** of the local culture of people. The term Ethnobotany was coined by **J.W. Harshberger** in 1895 to include the study of plants used by the primitive and aboriginal people. Though this discipline has existed for ages, ethnobotany emerged as a distinct academic branch of natural science in 20th century.

Aspects of ethnobotany

Ethnobotany has relevance with problems of nutrition, health care and life support system, faith in plants, cottage industries, economic upliftment, conservation of biodiversity and sustainable use of plant resources.

Importance of Ethnobotany

- It provides traditional uses of plant.
- It gives information about certain unknown and known useful plants.
- The ethnomedicinal data will serve as a useful source of information for the chemists, pharmacologists and practitioners of herbal medicine.
- Tribal communities utilize ethnomedicinal plant parts like bark, stem, roots, leaves, flower bud, flowers, fruits, seeds, oils, resins, dyes, gum for the treatment of diseases like diarrhoea, fever, headache, diabetes, jaundice, snakebites, leprosy, etc.

19.7 Astrobiology / Exobiology

Are we alone in the universe? If your answer is no, then how do you detect the existence of life in space? Astrobiology/exobiology is the science which looks for the presence of **extra terrestrial life in the universe**.

Astrobiology deals with the **origin, evolution** and **distribution of life in the universe** and to investigate the possibility of life in other world.

The major concept in astrobiology is the **habitable zone**. The theory explains that any **planets can support the existence of life**, if it fulfills two important criteria.

i It must have a right mass to retain an atmosphere.

ii It must have an orbit at just the right distance from its star (Sun) that it allows liquid water to exist. Thus, the distance need to be neither too hot or not too cold and is often referred as **Goldilock Zone for life**.

In our solar system 'Earth' is the only planet in the goldilock zone. Since, this zone varies at times as the star evolves, we know that Mars have also been habitable. The life on Mars are likely to be the creatures, we find in extreme environments on earth.

The organisms which live in extreme environmental conditions on earth are called **extremophiles.** Thus, within our own Solar System, there are many areas that are different from the Earth where it is probable to find the presence of life similar to extremophile bacteria.

10th_Science Unit-19.indd 281 06-12-2021 18:47:14